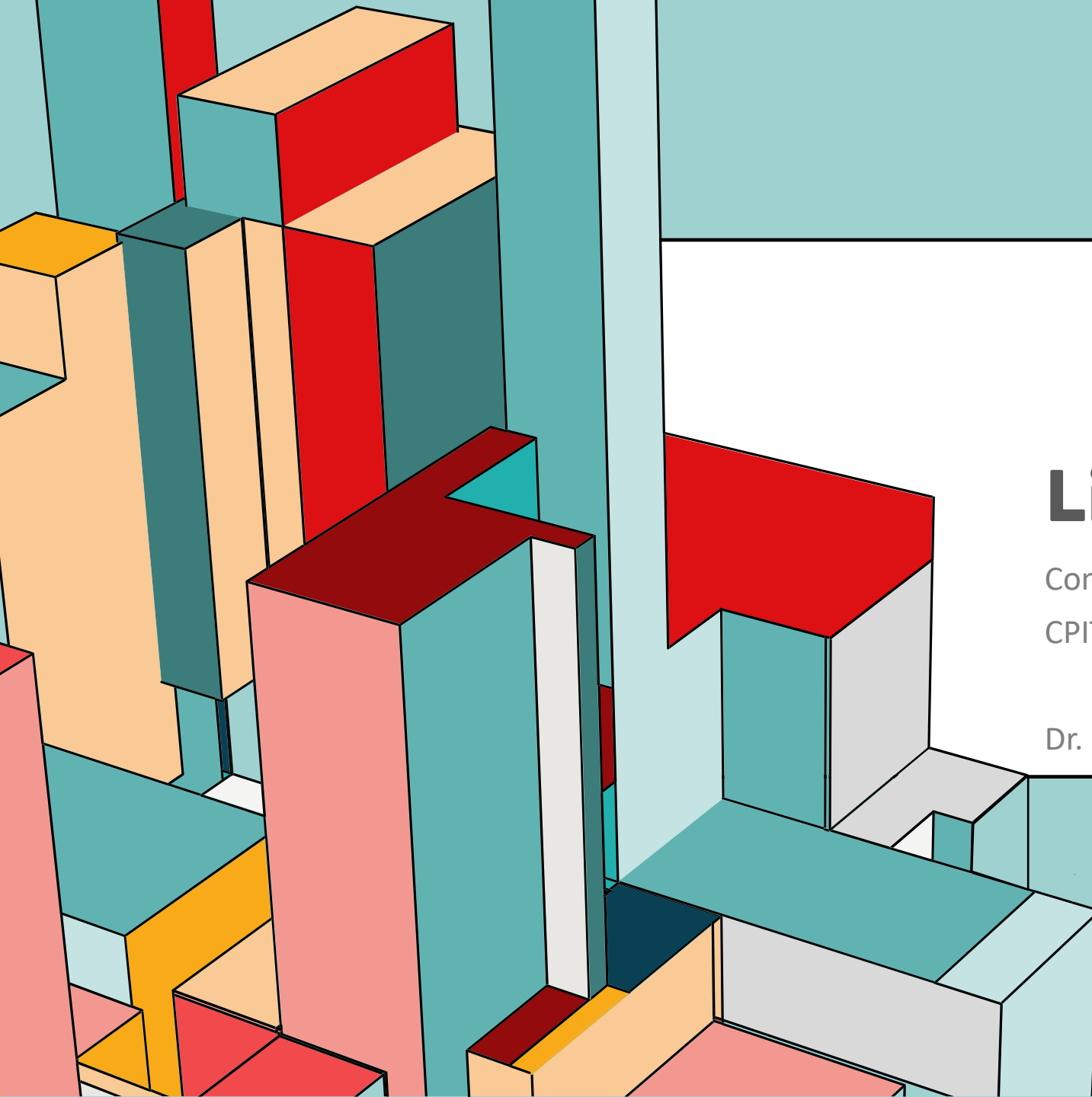




Lighting Effects

Computer Graphics

CPIT285

Dr. Rayed ALGhamdi

CONTENT

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Getting Started

02

Lighting Sources & Types

03

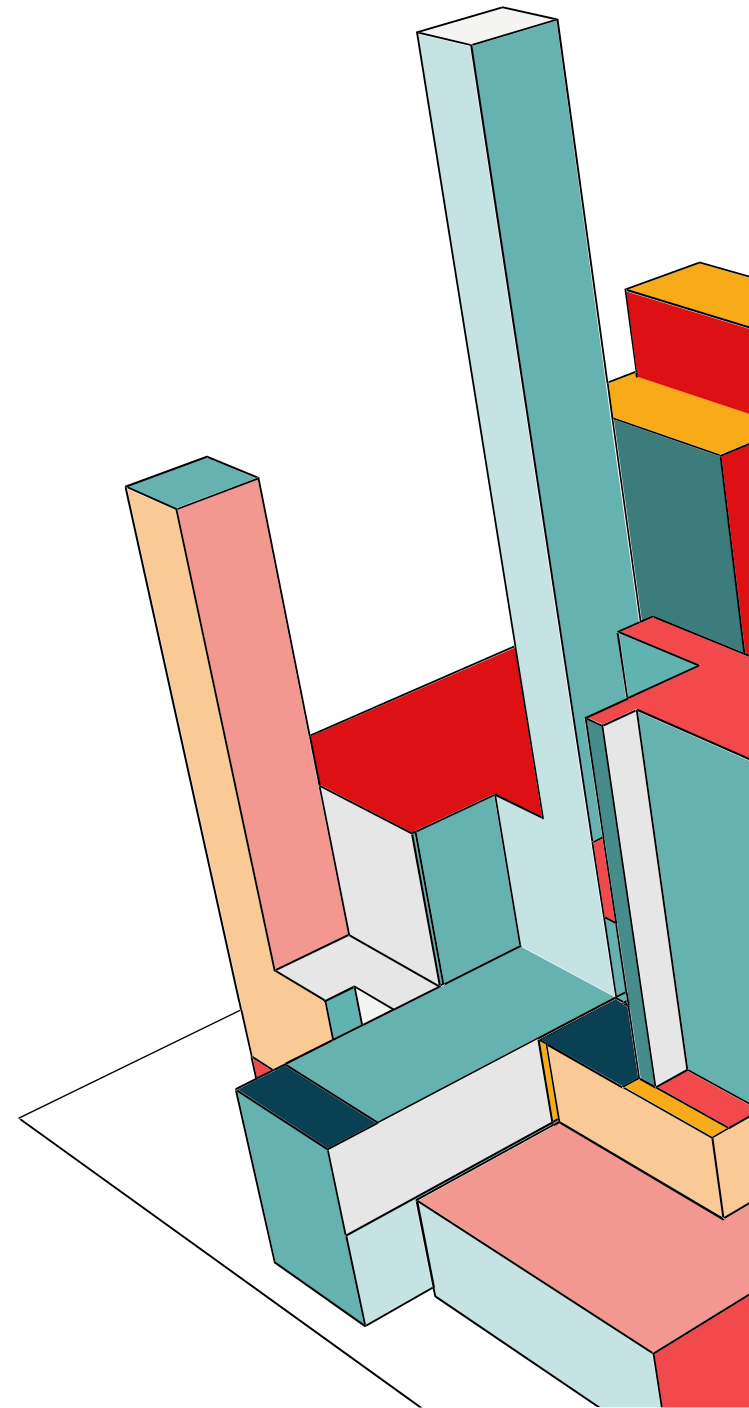
Impact of Lighting Color

04

Surface & Material Types

05

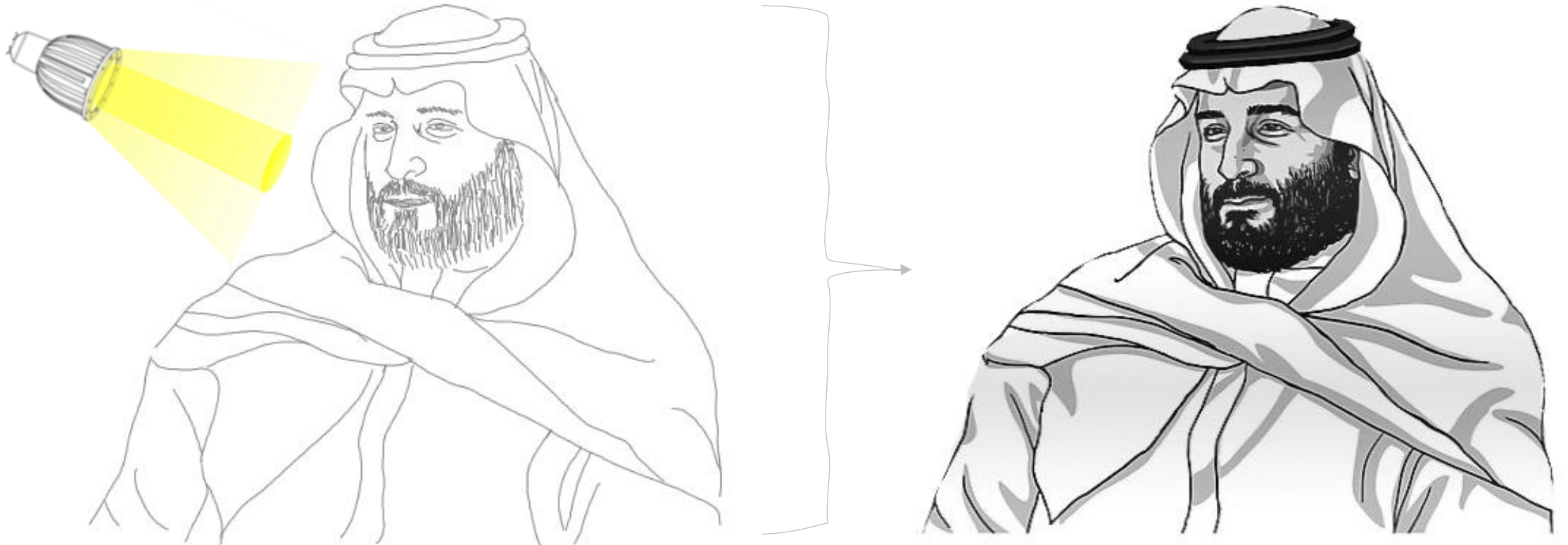
OpenGL Lighting Functions



LIGHTING EFFECTS

Getting Started

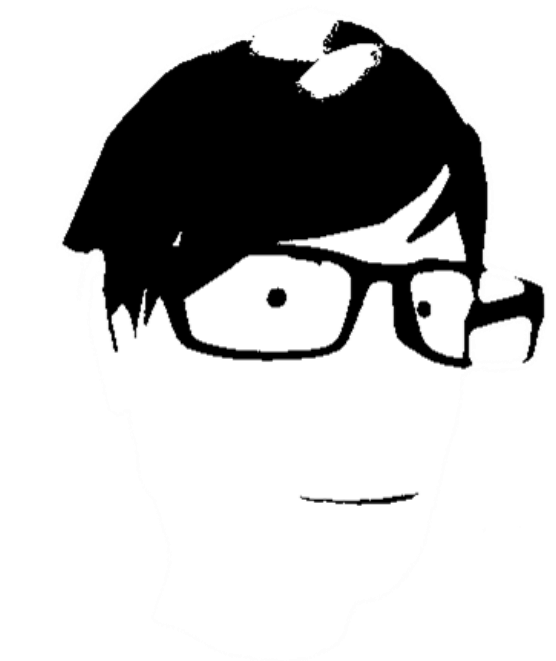
- How are lighting effects simulated in pencil drawing?



LIGHTING EFFECTS

Getting Started

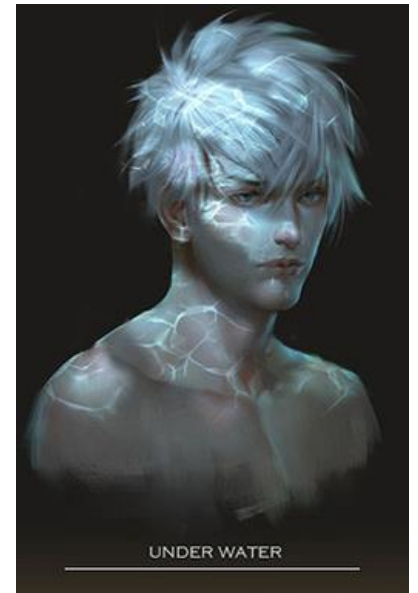
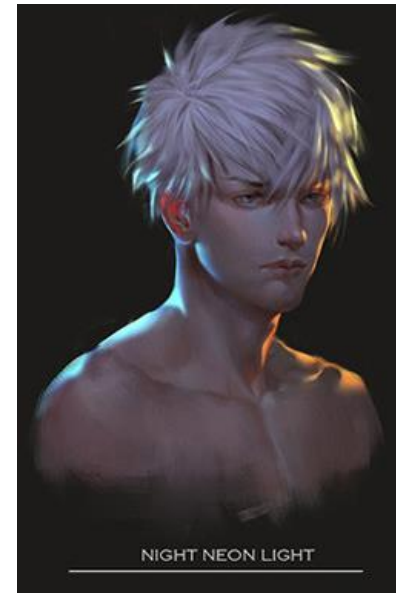
- How are lighting effects simulated in colored drawings?



LIGHTING EFFECTS

Getting Started

- No real light! It is all about playing with the RGB colors in the parts of the scenes...



```
globalAmbient [ ] = {0.1, 0.8, 0.3, 1.0};  
glLightModelfv (GL_LIGHT_MODEL_AMBIENT, globalAmbient);
```

LIGHTING EFFECTS

Getting Started

- Realistic displays of a scene are obtained by generating perspective projections of objects and applying natural lighting effects to the visible surfaces.
- A **lighting model** (also called illumination model and sometimes referred to as a shading model), is used to calculate the color of a lit position on the surface of an object.
- It is all about calculating the RGB colors for each part of an object that needs to simulate the lighting effects.
- These calculations of RGB colors to simulate lighting effects depend on a number of factors as indicated in the next slide.

LIGHTING EFFECTS

Lighting Factors

- The lighting model involves a number of factors:
 - Light source and attributes
(intensity, color, position, direction, shape, ...)
 - Object surface attributes
(material properties: color, reflectivity, transparency, ...)
 - Object orientation
Object position relative to the light and other objects
Interaction among lights and objects
 - viewing direction
Interaction between objects and eye



Lighting Sources & Types



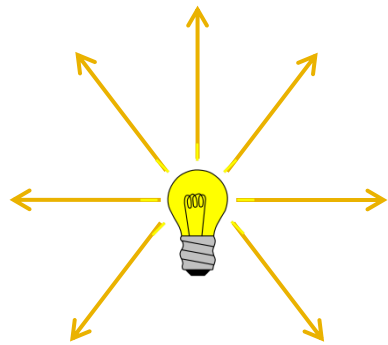
LIGHTING EFFECTS

Lighting Source

- Any object that is emitting radiant energy is a **light source** that contributes to the lighting effects for other objects in a scene.
- In most applications, and particularly for real-time graphics displays, a simple light-source model is used to avoid excessive computations.
- We assign light-emitting properties using a single value for each of the red, green, and blue (RGB) color components, which we can describe as the amount, or the “intensity,” of that color component.

LIGHTING EFFECTS

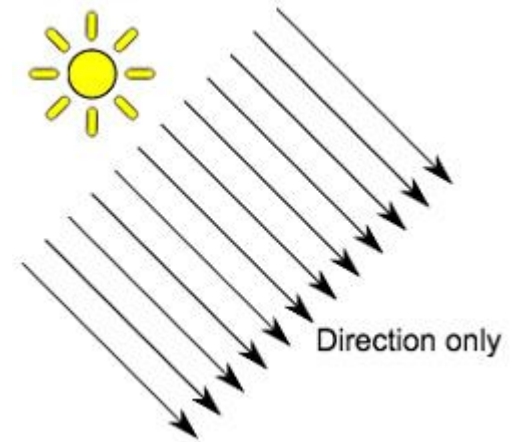
Lighting Source



Point Light

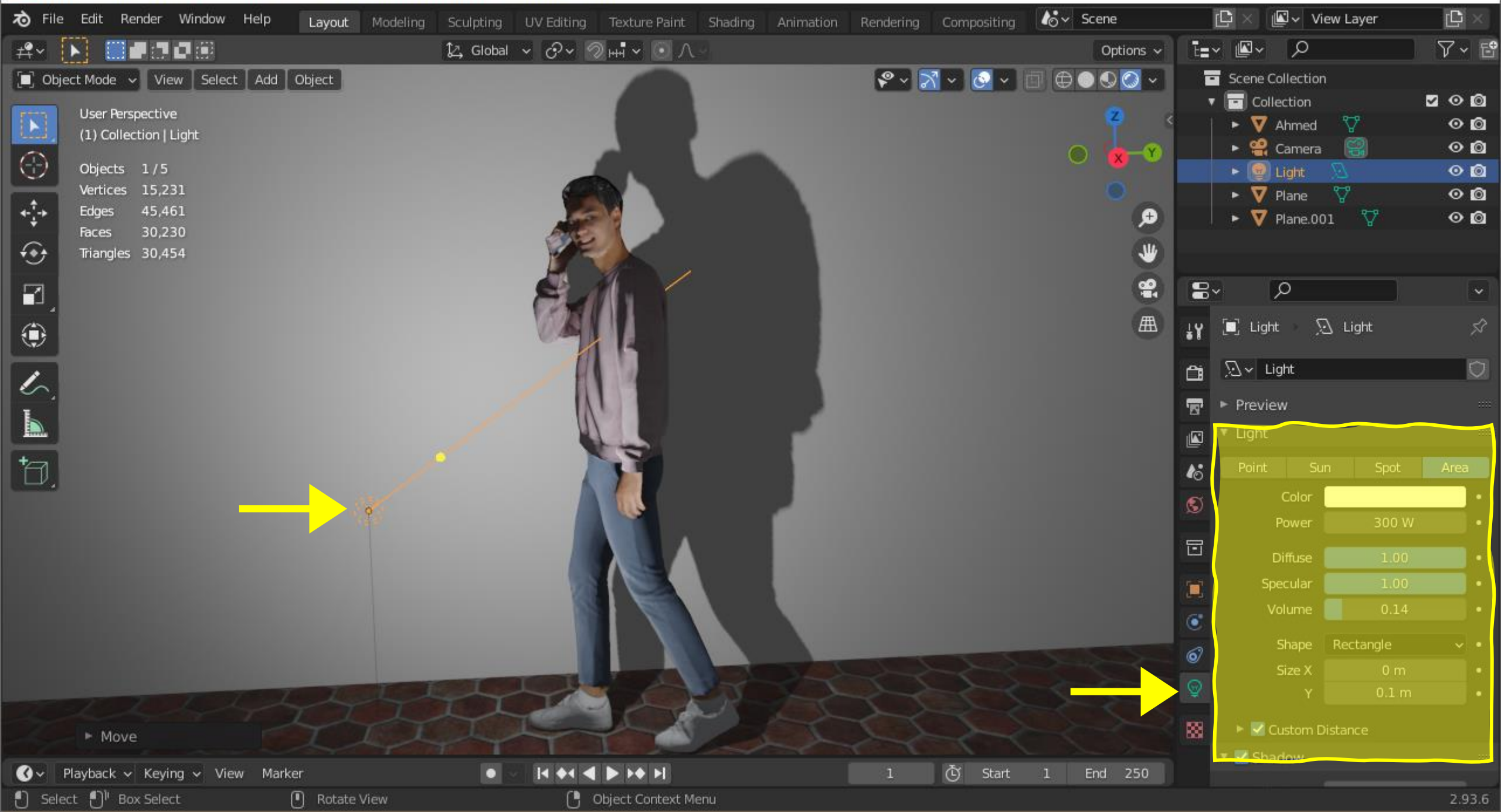


Spotlight



Directional Light

The light intensity can be independent or dependent on the distance between an object and the light source.

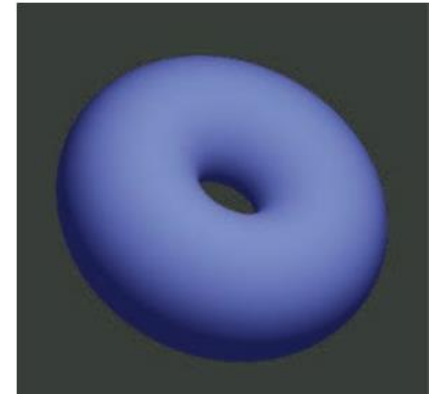


LIGHTING EFFECTS

Lighting Types

- Two of the most commonly used lighting models are the Lambert model and the Phong model, illustrated in the figure >>
- **Lambert model** uses a combination of ambient and diffuse lighting, particularly appropriate for simulating the appearance of matte or rough surfaces, where most of the light rays meeting the surface are scattered.
- **Phong model** uses ambient, diffuse, and specular lighting, and is particularly appropriate for reflective or shiny surfaces.

Lambert model



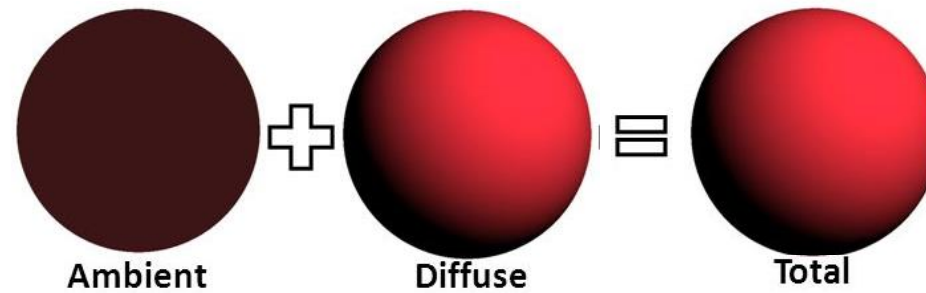
Phong model



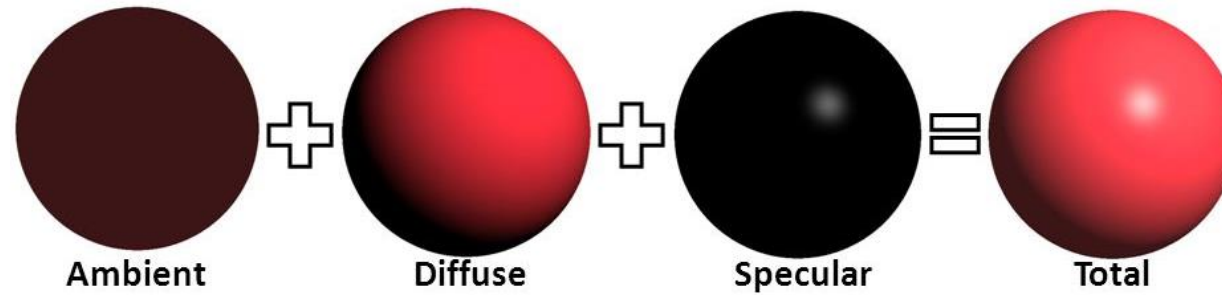
LIGHTING EFFECTS

Lighting Types

Lambert
Lighting
Model



Phong
Lighting
Model



LIGHTING EFFECTS

Lighting Types

Ambient

- The light that is scattered by the environment. It has no direction.
- Ambient light alone cannot communicate the complete representation of an object set in 3D space because all vertices are evenly lit by the same color and the object appears to be 2D.





Ambient light

LIGHTING EFFECTS

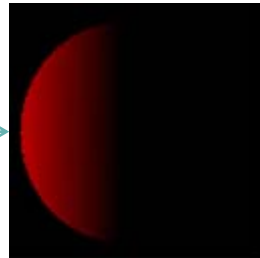
Lighting Types

Diffuse

- Diffuse light represents a directional light cast by a light source. It has a position in space and comes from a single direction.

Diffuse light

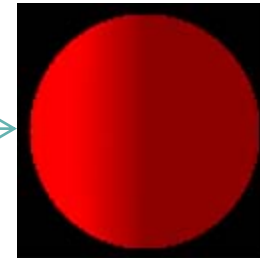
A diffuse light of red color is cast onto a black object defining its 3D shape.



Ambient

+

Diffuse





Ambient + Diffuse light

LIGHTING EFFECTS

Lighting Types

Specular

- Just like diffuse light, specular light is a directional type of light. It comes from one particular direction. The difference between the two is that specular light **reflects off the surface in a sharp and uniform way.**
- The rendering of specular light relies on the angle between the viewer and the light source.

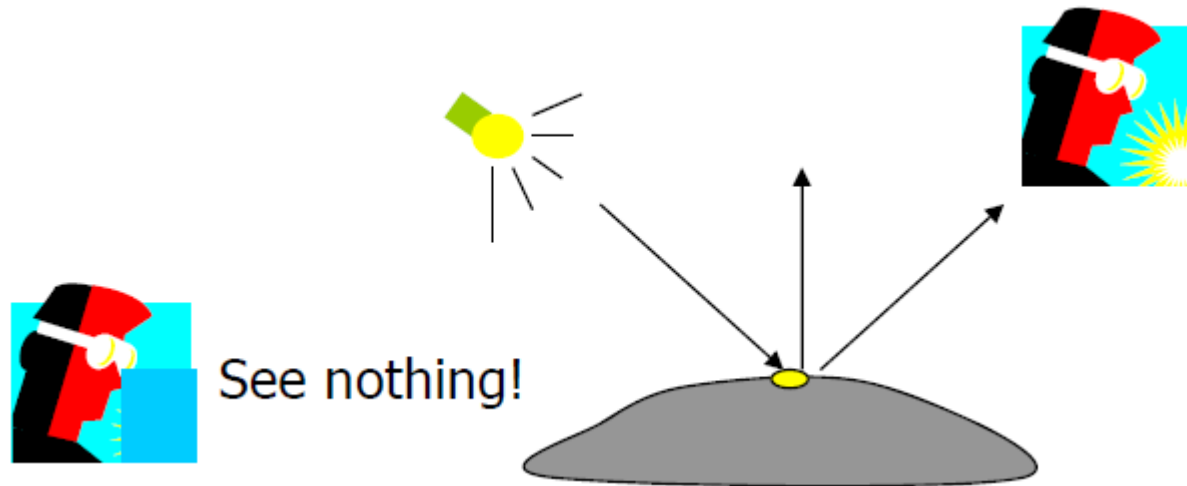


LIGHTING EFFECTS

Lighting Types

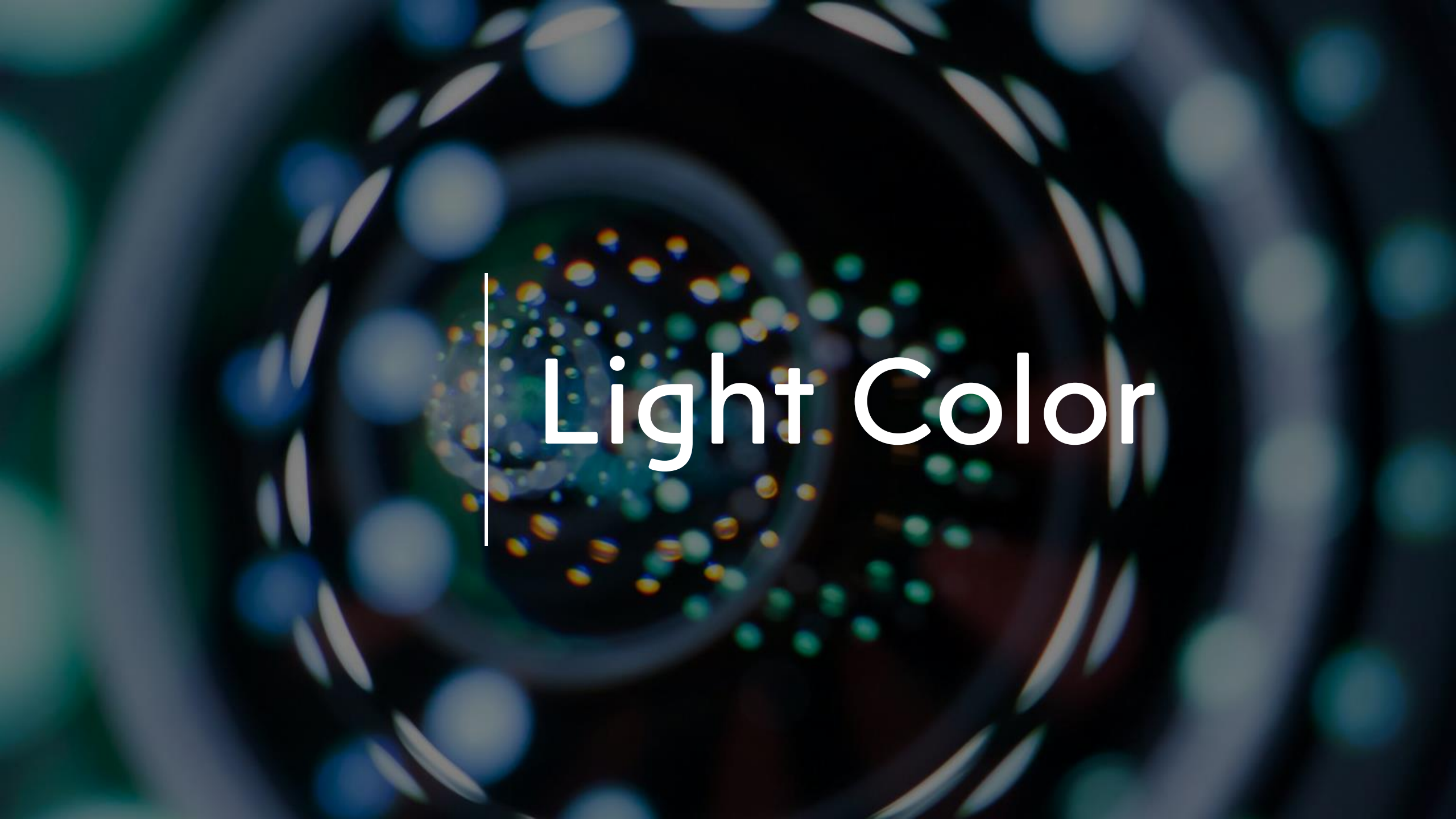
Specular

- How much reflection/bright spot you can see depends on where you are (viewing angle)





Ambient + Diffuse + Specular light

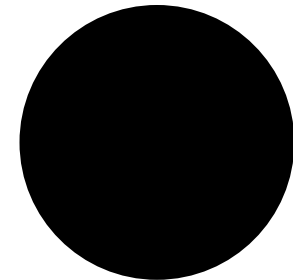
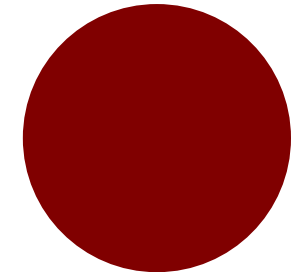
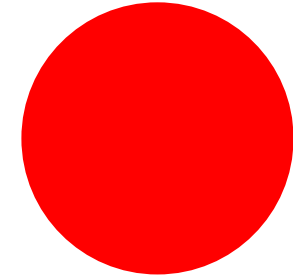
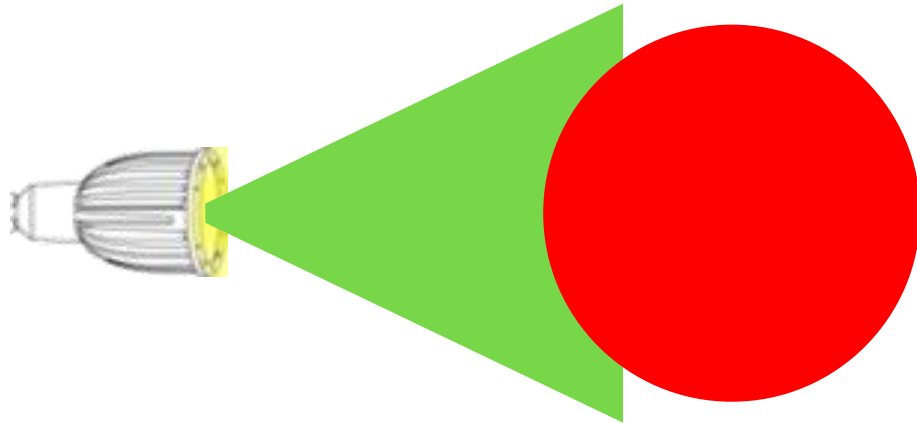


Light Color

LIGHTING EFFECTS

Light Color

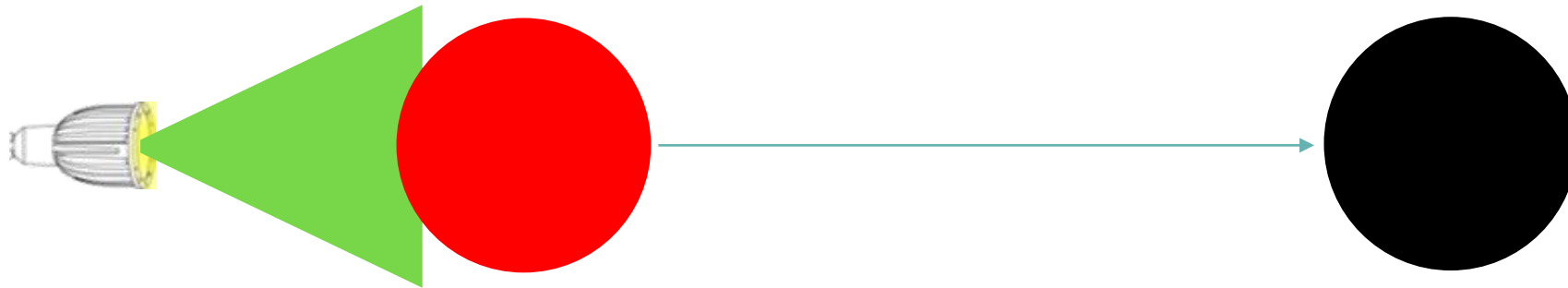
- When applying a green light to a red object, in which color will you view the object?



LIGHTING EFFECTS

Light Color

- When applying a green light to a red object, you see black color!
- The green light cannot reflect the red surface of the object.

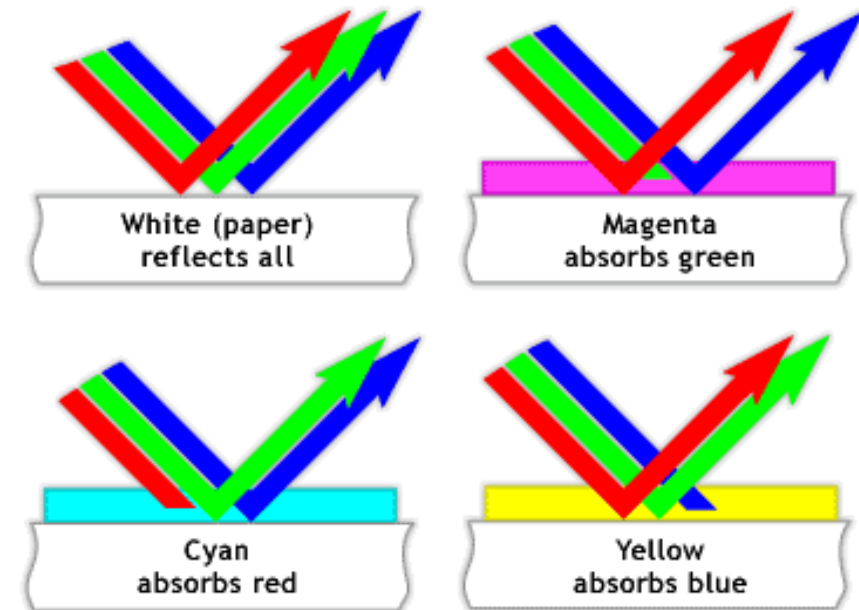
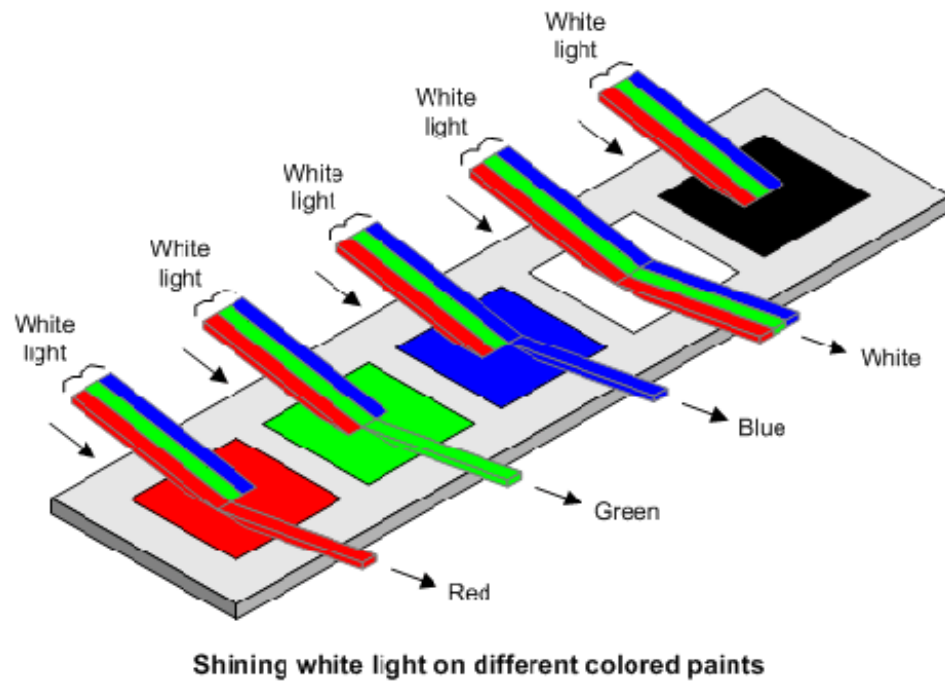


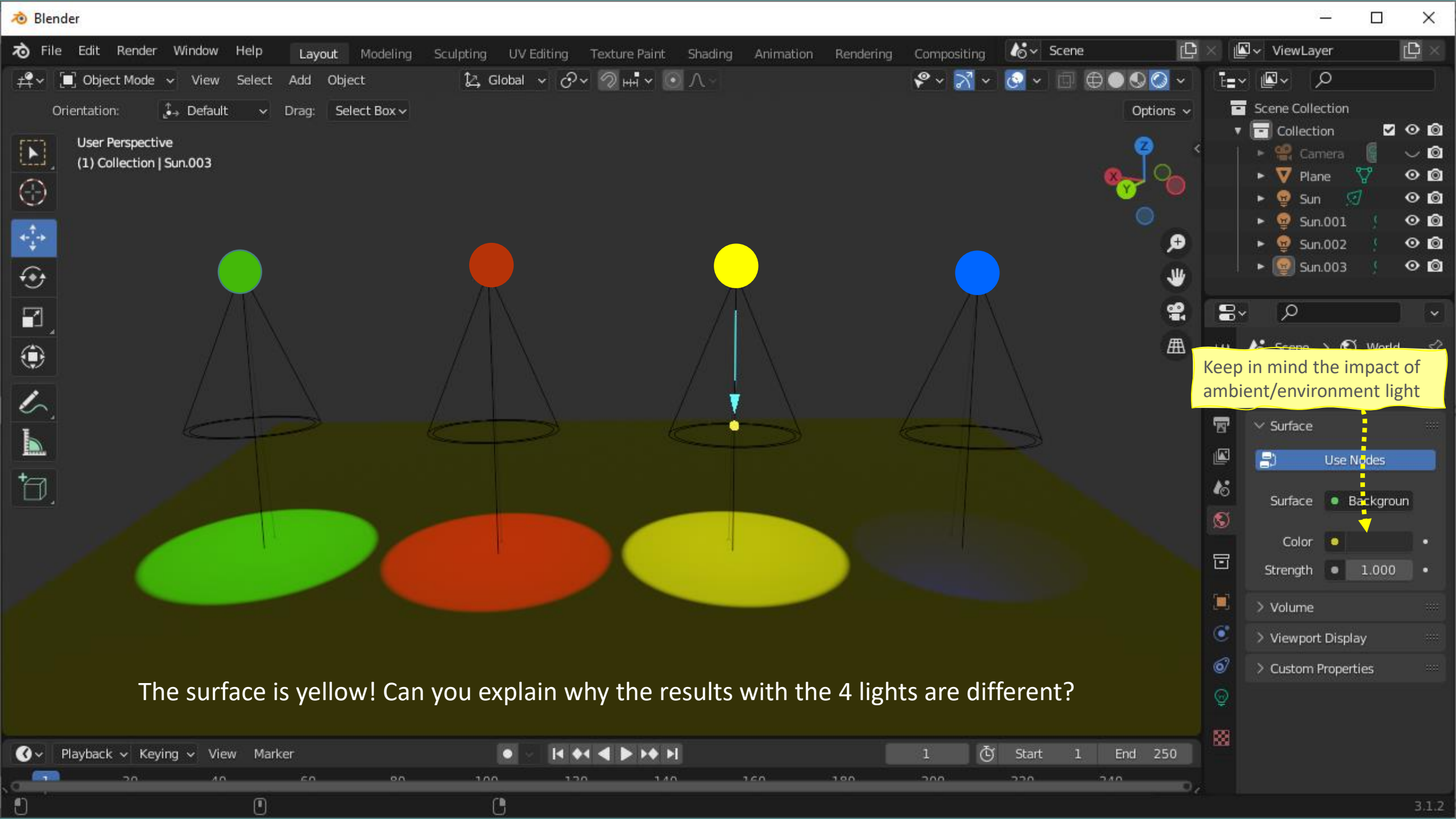
- This takes us to the concepts of colors reflection and absorption

LIGHTING EFFECTS

Light Color

- Reflecting and absorbing light





Keep in mind the impact of ambient/environment light

The surface is yellow! Can you explain why the results with the 4 lights are different?



R: 0
G: 255 / 1
B: 0

R: 255 / 1
G: 0
B: 0

R: 255 / 1
G: 255 / 1
B: 0

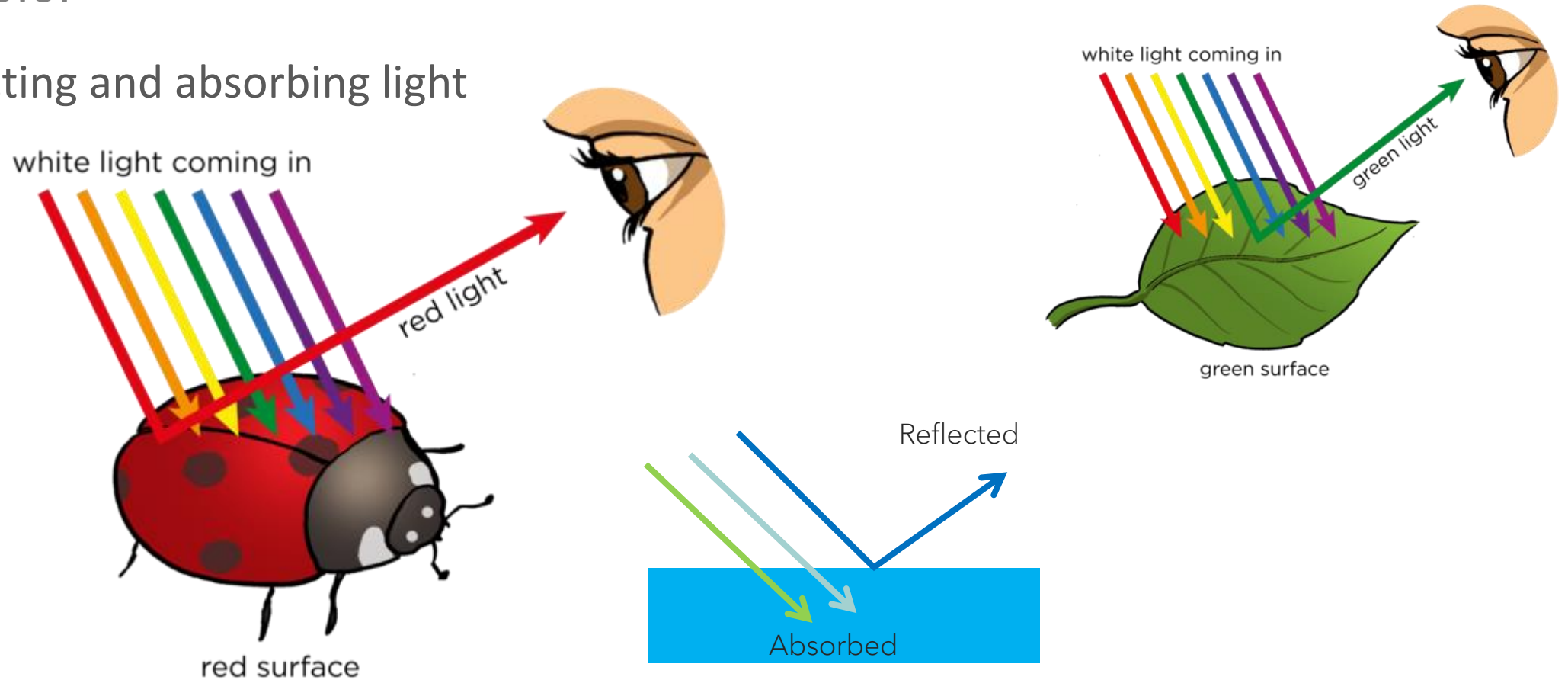
R: 0
G: 0
B: 255 / 1

The surface is Yellow (255,255,0) 

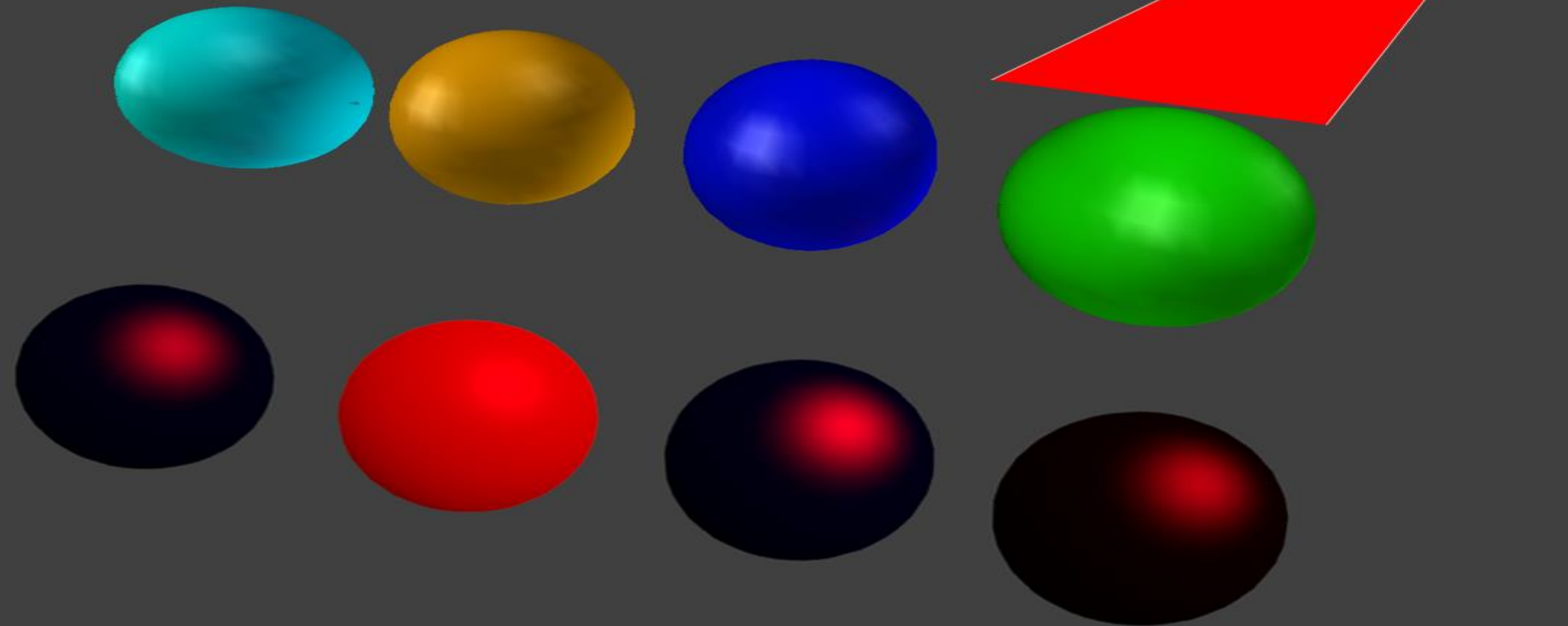
LIGHTING EFFECTS

Light Color

- Reflecting and absorbing light



Explain, why all balls turned black with a little red specular except one fully reflects the red light after rendering.



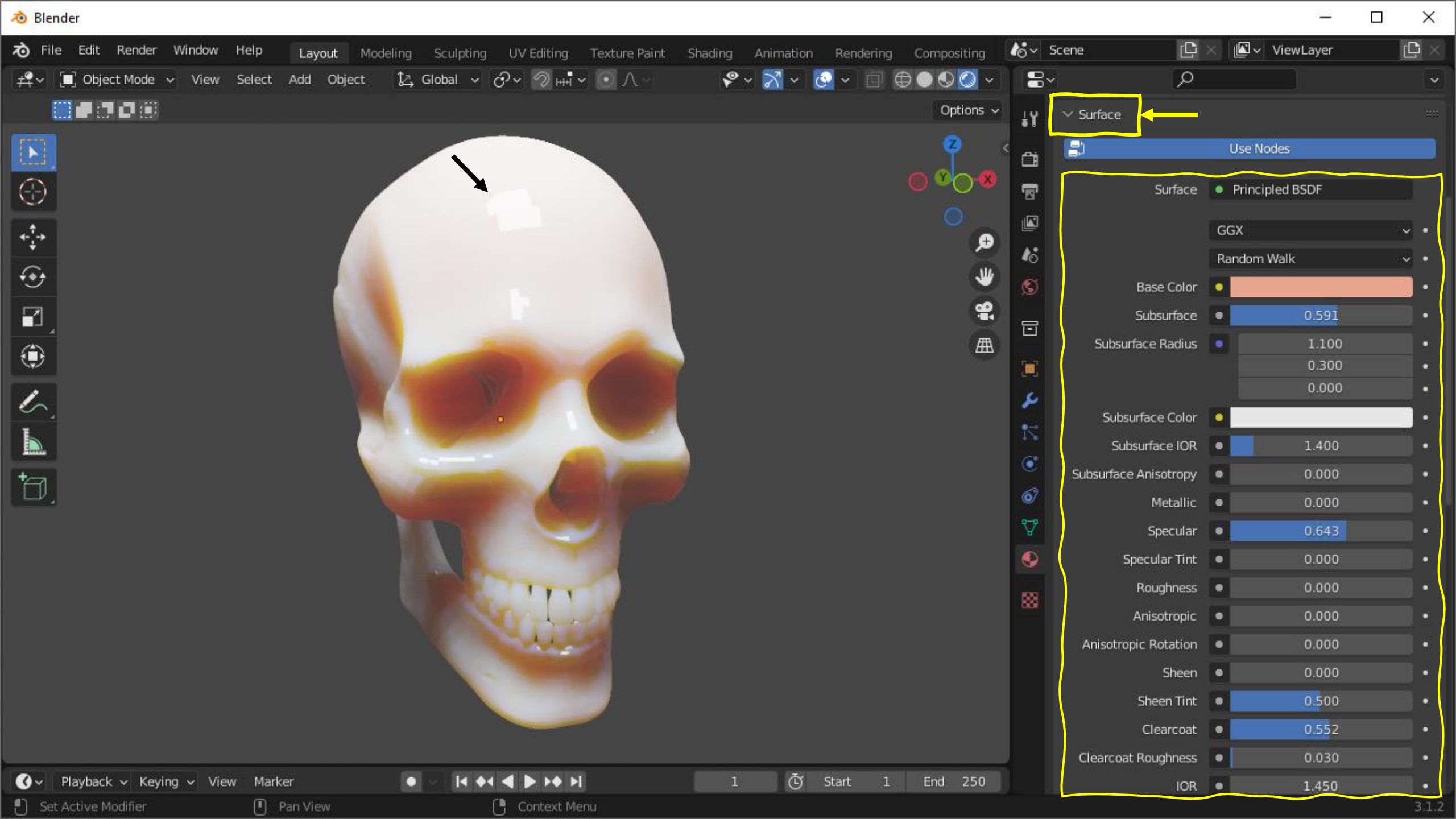
The background of the image is an abstract, high-contrast geometric pattern. It consists of a grid of squares, some of which are raised into a 3D effect, creating a series of steps or a staircase-like structure. The squares are in two colors: a dark, almost black, and a light, speckled grey. The lighting is dramatic, coming from the side, which casts long, dark shadows from the raised squares onto the adjacent squares, emphasizing the three-dimensional quality and the texture of the materials.

Surface & Material Types

SURFACE & MATERIAL TYPES

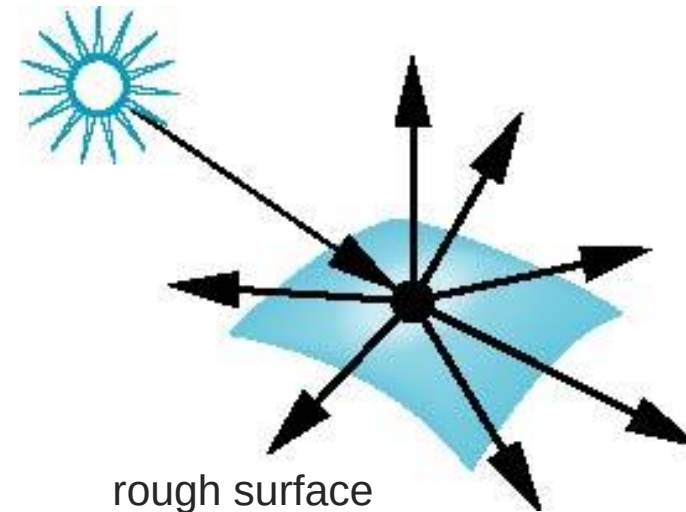
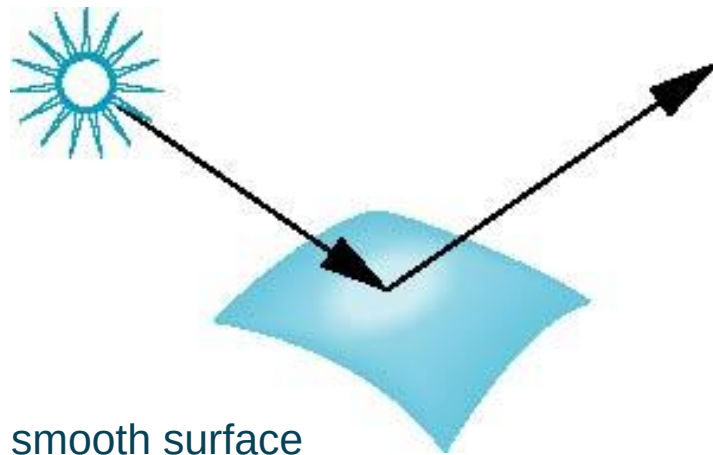
- As explained earlier; creating realistic lighting effects is not simply by adding a light source and that's it.
- Properties of lit objects need to be managed in order to reflect a real interaction.





SURFACE & MATERIAL TYPES

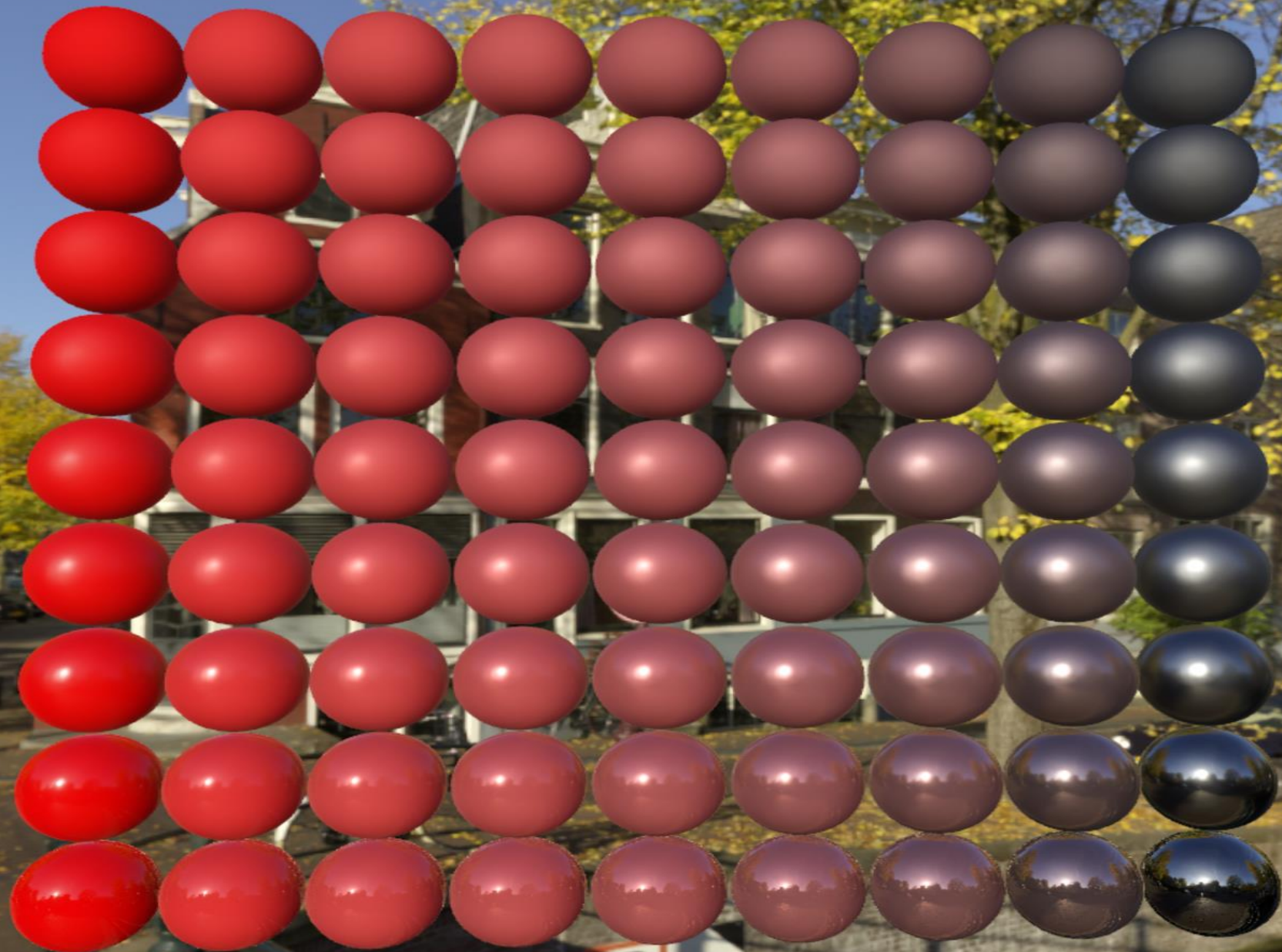
- The smoother a surface, the more reflected light is concentrated in the direction a perfect mirror would reflect light.
- A very rough surface scatters light in all directions.



SURFACE & MATERIAL TYPES



1.0 Roughness



0.0

Metalness

1.0









OpenGL Lighting Functions



OpenGL Lighting Functions

- A variety of routines are available in OpenGL for setting up point light sources, selecting surface-reflection coefficients, and choosing values for other parameters in the basic lighting model. In addition, we can simulate transparency, and objects can be displayed using either flat surface rendering or gouraud surface rendering.
- ...Consider what you learn in this course as an introduction to lighting. Let's work with very basic lighting process.

OpenGL Lighting Functions

Point Light Function

- Multiple point light sources can be included in an OpenGL scene description, and various properties, such as position, type, color, attenuation, and spotlight effects, are associated with each light source.
- We set a property value for a light source with the function

```
glLight* (lightName, lightProperty,  propertyValue);
```

OpenGL Lighting Functions

Point Light Function

```
glLight* (lightName, lightProperty,  propertyValue);
```

- Each light source is referenced with an identifier, and parameter `lightName` is assigned one of the OpenGL symbolic identifiers `GL_LIGHT0`, `GL_LIGHT1`, `GL_LIGHT2`,... `GL_LIGHT7`, although some implementations of OpenGL may allow more than 8 light sources.
- The parameter `lightProperty` must be assigned one of the 10 OpenGL symbolic property constants.

OpenGL Lighting Functions

Point Light Function

```
glLight* (lightName, lightProperty, propertyValue);
```

Constants	Default Value	Meaning
GL_AMBIENT	(0.0, 0.0, 0.0, 1.0)	RGBA for ambient item
GL_DIFFUSE	(1.0, 1.0, 1.0, 1.0)	RGBA for ambient item
GL_SPECULAR	(1.0, 1.0, 1.0, 1.0)	RGBA for ambient item
GL_POSITION	(0.0, 0.0, 1.0, 0.0)	Light position coordinates (x,y,z,w) w=0 →directional w<> 0 →positional

OpenGL Lighting Functions

Point Light Function

```
glLight* (lightName, lightProperty, propertyValue);
```

Constants	Default Value	Meaning
GL_SPOT_DIRECTION	(1.0, 0.0, 0.0)	(x, y, z) direction of spotlight
GL_SPOT_EXPONENT	0.0	spotlight exponent
GL_SPOT_CUTOFF	180.0	spotlight cutoff angle
GL_CONSTANT_ATTENUATION	1.0	constant attenuation factor
GL_LINEAR_ATTENUATION	0.0	linear attenuation factor
GL_QUADRATIC_ATTENUATION	0.0	quadratic attenuation factor

OpenGL Lighting Functions

Point Light Function

```
glLight* (lightName, lightProperty,  propertyValue);
```

```
// Create light components
```

```
Glfloat ambientLight[] = { 0.2, 0.2, 0.2, 1.0 };
```

```
Glfloat diffuseLight[] = { 0.8, 0.8, 0.8, 1.0 };
```

```
Glfloat specularLight[] = { 0.5, 0.5, 0.5, 1.0 };
```

```
Glfloat position[] = { -1.5, 1.0, -4.0, 1.0 };
```

```
// Assign created components to GL_LIGHT0
```

```
glLightfv(GL_LIGHT0, GL_AMBIENT, ambientLight);
```

```
glLightfv(GL_LIGHT0, GL_DIFFUSE, diffuseLight);
```

```
glLightfv(GL_LIGHT0, GL_SPECULAR, specularLight);
```

```
glLightfv(GL_LIGHT0, GL_POSITION, position);
```

OpenGL Lighting Functions

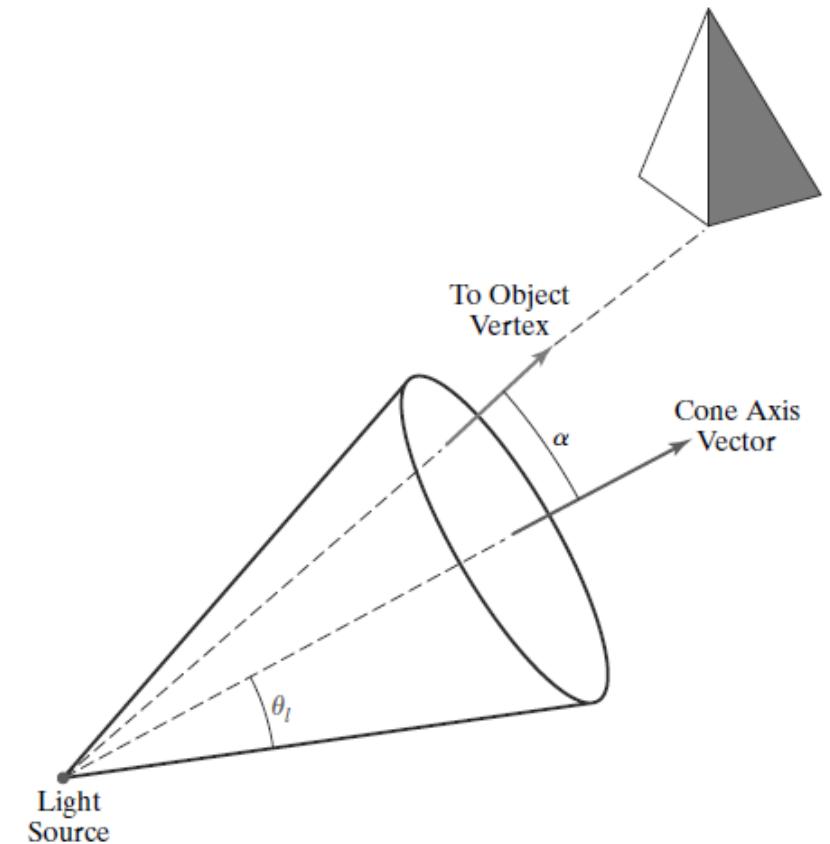
Point Light Function

- After all properties have been assigned to a light source, we turn on that light with the command:
`glEnable (lightName);`
- However, we also need to activate the OpenGL lighting routines, and we do that with the command:
`glEnable (GL_LIGHTING);`
- To turnoff a light source, use the command:
`glDisable (lightName);`

OpenGL Lighting Functions

Spotlight Function

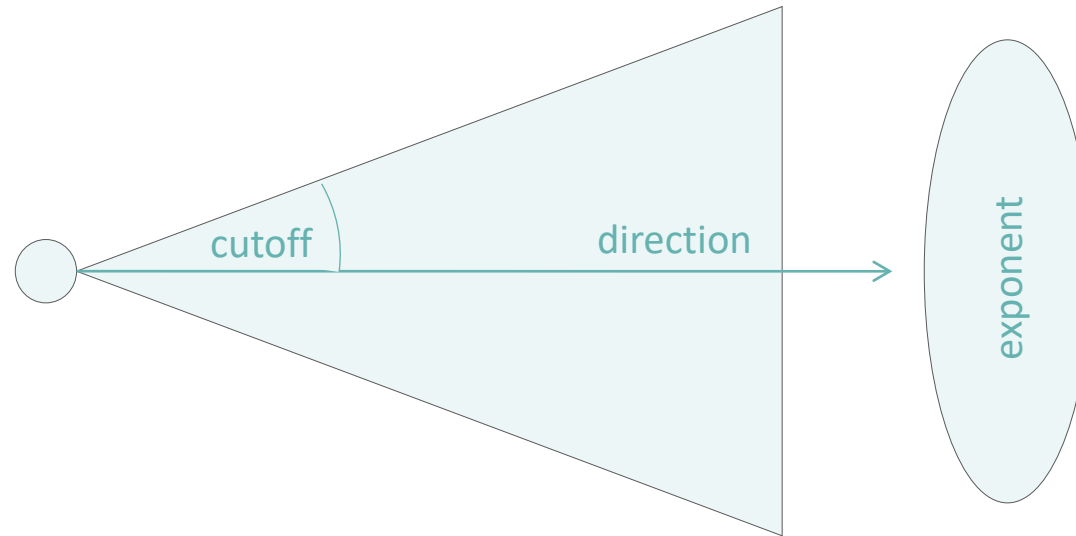
- Spotlight source limits the light to a cone-shaped region of space.
- A circular cone of light emitted from an OpenGL light source. The angular extent of the light cone, measured from the cone axis, is ϑ_l , and the angle from the axis to an object direction vector is labeled as α .
- If $\alpha > \vartheta_l$, the object is outside the light-source cone, and the object is not illuminated by this light source.



OpenGL Lighting Functions

Spotlight Function

- Three components need to be specified:
 - `GL_SPOT_DIRECTION`
 - `GL_SPOT_CUTOFF`
 - `GL_SPOT_EXPONENT`



OpenGL Lighting Functions

Spotlight Function

- If we do not specify a direction for a light source, the default direction is parallel to the negative z axis; that is, (0.0, 0.0, -1.0). Also, the default cone angle is 180° and the default attenuation exponent is 0. Thus, the default is a point light source that radiates in all directions, with no angular attenuation.

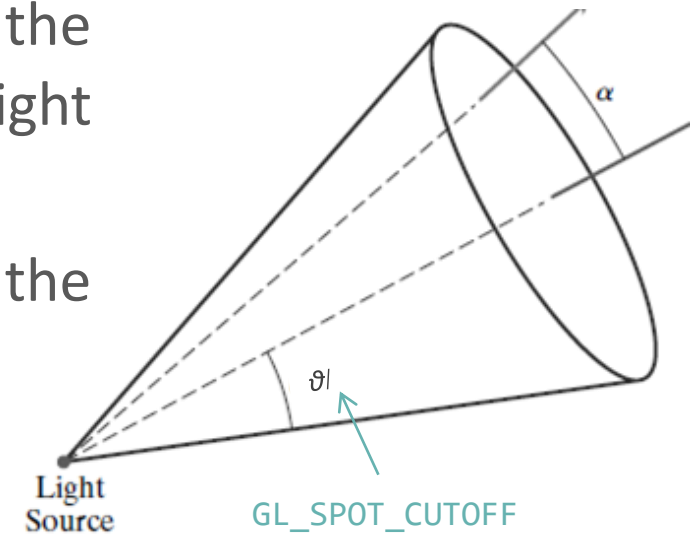
Constants	Default Value	Meaning
GL_SPOT_DIRECTION	(0.0, 0.0, -1.0)	(x, y, z) direction of spotlight
GL_SPOT_EXPONENT	0.0	spotlight exponent
GL_SPOT_CUTOFF	180.0	spotlight cutoff angle

OpenGL Lighting Functions

Spotlight Function

- To specify the angle between the axis of the cone and a ray along the edge of the cone, use the `GL_SPOT_CUTOFF` parameter.
- By default, the spotlight feature is disabled because the `GL_SPOT_CUTOFF` parameter is 180.0. This value means that light is emitted in all directions.
- The value for `GL_SPOT_CUTOFF` is restricted to being within the range [0.0,90.0]

```
glLightf(GL_LIGHT0, GL_SPOT_CUTOFF, 45.0);
```



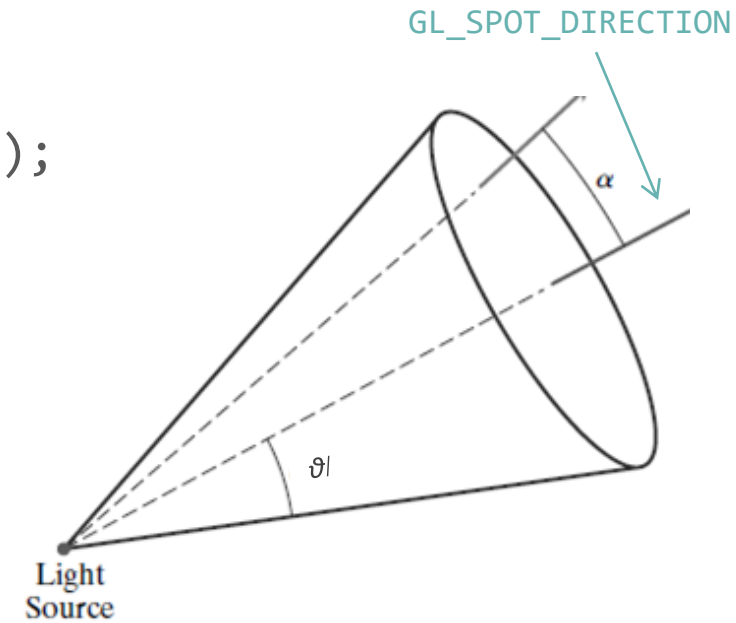
OpenGL Lighting Functions

Spotlight Function

- To specify a spotlight's direction:

```
GLfloat spot_direction[] = { -1.0, -1.0, 0.0 };  
glLightfv(GL_LIGHT0, GL_SPOT_DIRECTION, spot_direction);
```

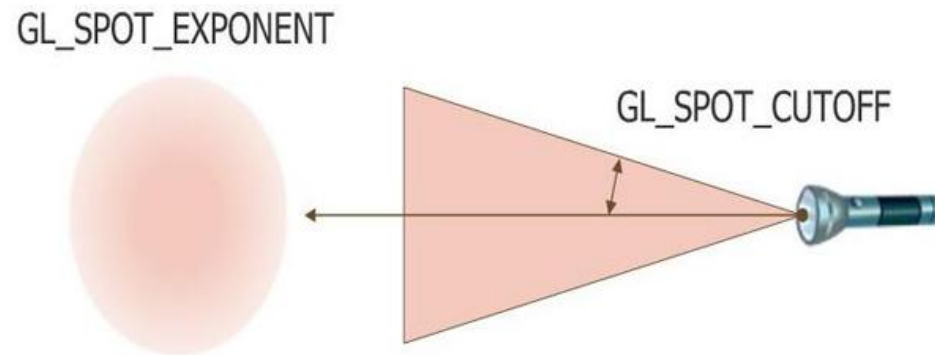
- By default, the direction is (0.0, 0.0, -1.0)
- When specifying the direction, make it toward the object coordinates.



OpenGL Lighting Functions

Spotlight Function

- To control the intensity distribution of the light within the cone of the spotlight, `GL_SPOT_EXPONENT` parameter is set.
- The parameter of `GL_SPOT_EXPONENT` is zero by default.
- It takes values in the range of 0 to 128.
- Higher spot exponents result in a more focused light source.



OpenGL Lighting Functions

Spotlight Function

- To control the intensity distribution of the light within the cone of the spotlight,
- E.g. The following statements set the directional effects for light source 3 so that the cone axis is in the positive x direction, the cone angle θ is 30° , and the attenuation exponent is 2.5:

```
GLfloat dirVector [ ] = {1.0, 0.0, 0.0};  
  
glLightfv (GL_LIGHT3, GL_SPOT_DIRECTION, dirVector);  
glLightf (GL_LIGHT3, GL_SPOT_CUTOFF, 30.0);  
glLightf (GL_LIGHT3, GL_SPOT_EXPONENT, 2.5);
```

OpenGL Lighting Functions

Surface-Property Function

- This section describes how to define the material properties of the objects in the scene: the ambient, diffuse, and specular colors, the shininess, and the color of any emitted light.
- Most of the material properties are conceptually similar to ones you've already used to create light sources.
- The mechanism for setting them is similar, except that the command used is called `glMaterial*()`

OpenGL Lighting Functions

Surface-Property Function

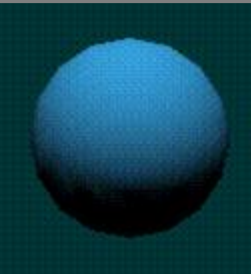
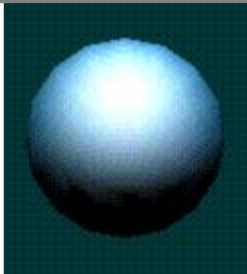
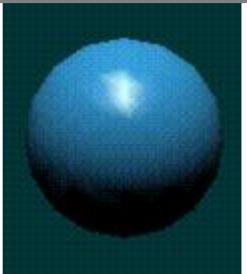
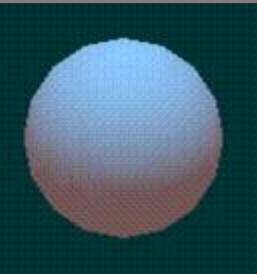
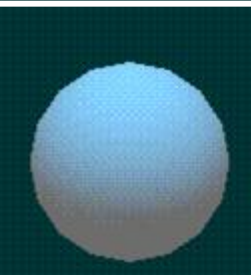
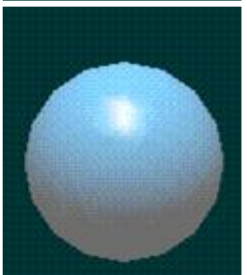
```
glMaterial* (surfFace, surfProperty, propertyValue);
```



GL_FRONT
GL_BACK
GL_FRONT_AND_BACK

Constant	Default Value	Meaning
GL_AMBIENT	(1.0 ,0.2 ,0.2 ,0.2)	ambient color of material
GL_DIFFUSE	(1.0 ,0.8 ,0.8 ,0.8)	diffuse color of material
GL_AMBIENT_AND_DIFFUSE		ambient and diffuse color of material
GL_SPECULAR	(1.0 ,0.0 ,0.0 ,0.0)	specular color of material
GL_SHININESS	0.0	specular exponent
GL_EMISSION	(1.0 ,0.0 ,0.0 ,0.0)	emissive color of material
GL_COLOR_INDEXES	(0,1,1)	ambient, diffuse & specular color indices

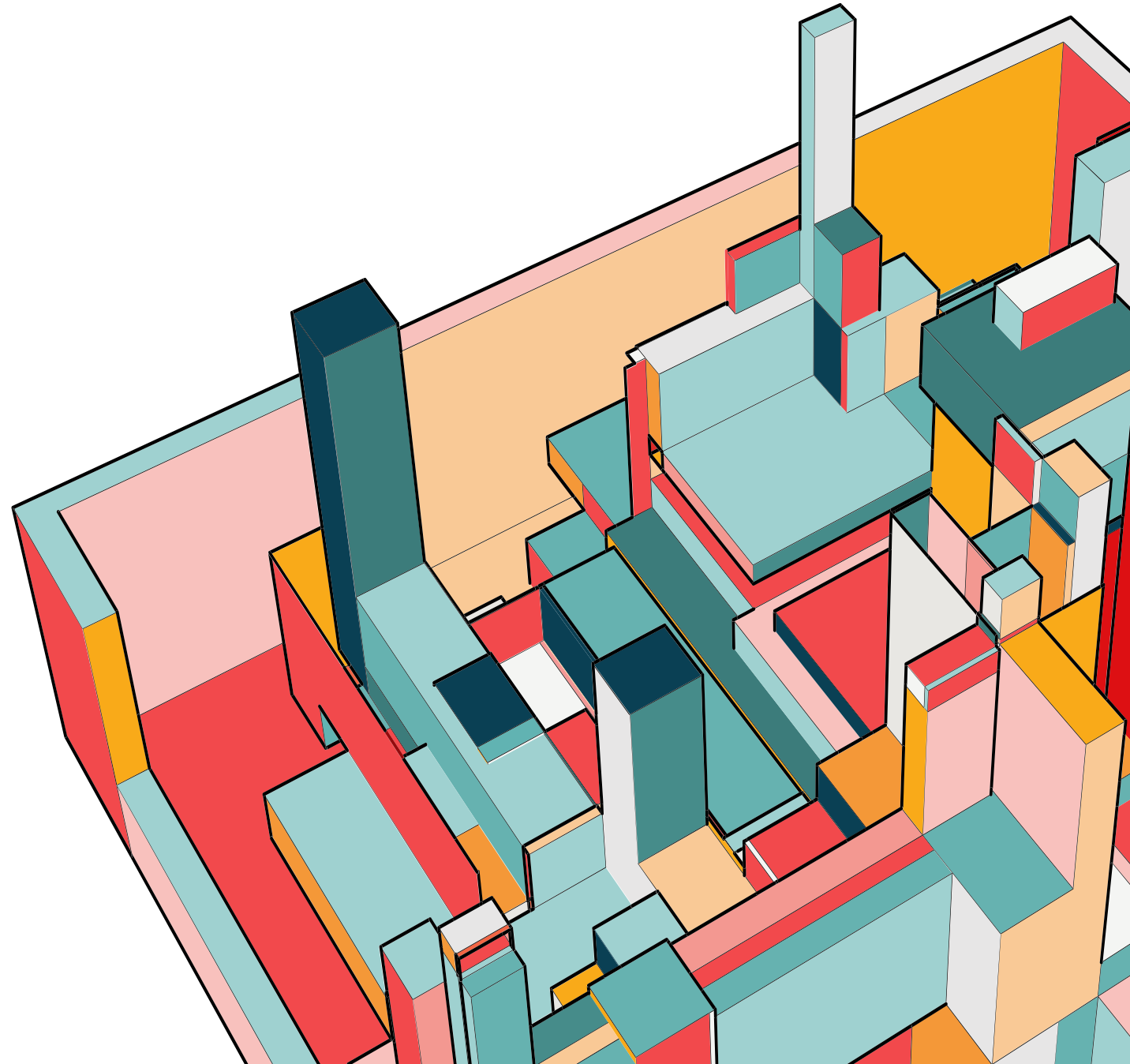
Surface-Property Function

	Blue diffuse /no specular	White specular + low shininess	High shininess exponent	Blue diffuse + emissive component
No ambient reflection				
Grey ambient reflection				
Blue ambient reflection				

100% CONFIDENT

Your journey in this course is
enjoyable and beneficial!

Enjoy learning CPIT285!



THANK YOU

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